



Method 1: Brute Force.

① Change-of-Variable:

$$\text{Let } z_x = \frac{x - \mu_x}{\sigma_x} \quad z_y = \frac{y - \mu_y}{\sigma_y}$$

Then, we can reorganize the joint dist. to be

$$\begin{aligned} f(x, y) &= \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}[z_x^2 - 2\rho z_x z_y + z_y^2]\right\} \\ &= \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}[z_y^2 - 2\rho z_x z_y + \rho^2 z_x^2 - \rho^2 z_x^2 + z_x^2]\right\} \\ &= \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}(z_y - \rho z_x)^2 - \frac{1}{2}z_x^2\right\} \\ &= \frac{1}{\sqrt{2\pi}\sigma_y^2(1-\rho^2)} \exp\left\{-\frac{1}{2(1-\rho^2)}(z_y - \rho z_x)^2\right\} \frac{1}{\sqrt{2\pi}\sigma_x^2} \exp\left\{-\frac{1}{2}z_x^2\right\} \\ &= h(x, y) g(x) \end{aligned}$$

Let's focus on $h(x, y)$.

$$\begin{aligned} (z_y - \rho z_x)^2 &= \left(\frac{y - \mu_y}{\sigma_y} - \rho \frac{x - \mu_x}{\sigma_x}\right)^2 \\ &= \frac{1}{\sigma_y^2} \left(y - \mu_y - \rho x \frac{\sigma_y}{\sigma_x} + \rho \mu_x \frac{\sigma_y}{\sigma_x}\right)^2 \end{aligned}$$

Therefore,

$$h(x, y) = \frac{1}{\sqrt{2\pi}\sigma_y^2(1-\rho^2)} \exp\left\{-\frac{1}{2(1-\rho^2)\sigma_y^2} \left(y - \mu_y - \rho x \frac{\sigma_y}{\sigma_x} + \rho \mu_x \frac{\sigma_y}{\sigma_x}\right)^2\right\}$$

② Marginal Distribution of x :

$$\begin{aligned} f_x(x) &= \int p(x, y) dy \\ &= \int h(x, y) g(x) dy \\ &= g(x) \int \frac{1}{\sqrt{2\pi}\sigma_y^2(1-\rho^2)} \exp\left\{-\frac{1}{2(1-\rho^2)\sigma_y^2} \left(y - \mu_y - \rho x \frac{\sigma_y}{\sigma_x} + \rho \mu_x \frac{\sigma_y}{\sigma_x}\right)^2\right\} dy. \end{aligned}$$

Note that $h(x, y)$ is a valid normal density for y given x which means $\int h(x, y) dy = 1$

Therefore, $f_X(x) = g(x)$

② Conditional Distribution of $Y|X$

$$f_{Y|X}(y) = \frac{f(x, y)}{f_X(x)} = \frac{h(x, y) g(x)}{f_X(x)} \\ = h(x, y)$$

$$= \frac{1}{\sqrt{2\pi\sigma_Y^2(1-\rho^2)}} \exp\left\{-\frac{1}{2(1-\rho^2)\sigma_Y^2} \left(y - \mu_Y - \rho x \frac{\sigma_Y}{\sigma_X} + \rho \mu_X \frac{\sigma_Y}{\sigma_X}\right)^2\right\}$$

Compare with $N(\mu, \sigma^2)$ density,

$$\frac{1}{\sqrt{2\pi}\sigma^2} \exp\left\{-\frac{1}{2\sigma^2} (x - \mu)^2\right\}$$

Therefore, $Y|X \sim N(\mu_X, \sigma_X^2)$

$$\text{where } \sigma_X^2 = \sigma_Y^2(1-\rho^2) \quad \mu_X = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)$$

Method 2: Matrix Algebra + Construction of independent dist.

From the matrix representation of bivariate normal, such $f(x, y)$ implies

$$\begin{pmatrix} x \\ y \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}\right)$$

One can check by computing it through $N(\vec{\mu}, \Sigma)$ formula:

$$\frac{1}{(2\pi)^{p/2} |\Sigma|} \exp\left\{-\frac{1}{2} (\vec{x} - \vec{\mu})^T \Sigma^{-1} (\vec{x} - \vec{\mu})\right\}$$

Then, we want to utilize the prop. of normal dist. through linear combination and find $Z = Y - \rho X = [-\rho \ 1] \begin{bmatrix} X \\ Y \end{bmatrix}$ that is independent of X .

The covariance of (X, Z) would be.

$$\begin{bmatrix} X \\ Z \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ -\rho & 1 \end{bmatrix}}_A \begin{bmatrix} X \\ Y \end{bmatrix}$$

$$\begin{aligned}
\Sigma_{XZ} &= A \Sigma A^T \\
&= \begin{bmatrix} 1 & 0 \\ -c & 1 \end{bmatrix} \begin{bmatrix} \sigma_X^2 & \rho \sigma_X \sigma_Y \\ \rho \sigma_X \sigma_Y & \sigma_Y^2 \end{bmatrix} \begin{bmatrix} 1 & -c \\ 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} \sigma_X^2 & \rho \sigma_X \sigma_Y \\ \rho \sigma_X \sigma_Y - c \sigma_X^2 & \sigma_Y^2 - c \rho \sigma_X \sigma_Y \end{bmatrix} \begin{bmatrix} 1 & -c \\ 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} \sigma_X^2 & -c \sigma_X^2 + \rho \sigma_X \sigma_Y \\ \rho \sigma_X \sigma_Y - c \sigma_X^2 & \sigma_Y^2 - 2c \rho \sigma_X \sigma_Y + c^2 \sigma_X^2 \end{bmatrix}
\end{aligned}$$

Hence, to make $Z \perp X$, it is equivalent to covariance equals 0,
 $\rho \sigma_X \sigma_Y - c \sigma_X^2 = 0 \Rightarrow c = \frac{\rho \sigma_Y}{\sigma_X}$

And then, the mean and variance of Z is

$$E Z = E Y - c E X = \mu_Y - \frac{\rho \sigma_Y}{\sigma_X} \mu_X$$

$$\begin{aligned}
\text{Var}(Z) &= \text{Var}(Y) + c^2 \text{Var}(X) - 2c \text{Cov}(X, Y) \\
&= \sigma_Y^2 + \frac{\rho^2 \sigma_Y^2}{\sigma_X^2} \sigma_X^2 - 2 \rho \frac{\sigma_Y}{\sigma_X} \rho \sigma_X \sigma_Y \\
&= (1 - \rho^2) \sigma_Y^2
\end{aligned}$$

By independence,

$$Y - \rho \frac{\sigma_Y}{\sigma_X} X \mid X \sim N\left(\mu_Y - \frac{\rho \sigma_X}{\sigma_Y} \mu_X, (1 - \rho^2) \sigma_Y^2\right)$$

$$\Rightarrow Y \mid X \sim N\left(\mu_Y - \frac{\rho \sigma_X}{\sigma_Y} \mu_X + \rho \frac{\sigma_X}{\sigma_Y} X, (1 - \rho^2) \sigma_Y^2\right)$$

Since we condition on X , and X is treated as a constant.

Connection with OLS

Suppose we observe $(x_1, y_1), \dots, (x_n, y_n)$ generated from the bivariate normal distribution.

Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ the fitted SLR

It is known that

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

→ Sample Version

From the above derivation, we know,

$$EY = \underbrace{\mu_Y - \rho \frac{\sigma_Y}{\sigma_X} \mu_X}_{\beta_0} + \underbrace{\rho \frac{\sigma_Y}{\sigma_X}}_{\beta_1} X$$

Therefore, $\beta_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{\rho \sigma_X \sigma_Y}{\sigma_X^2} = \rho \frac{\sigma_Y}{\sigma_X} \rightarrow \text{Population Version.}$

Computing the mean of $Y|X$ is equivalent to the linear regression of Y given X .